

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Tech. Examination (ME/CL/EE)

MAY 2014

EE 103 Basics of Electronics & Electrical Engineering

Date: 22.05.2014, Thursday

Time: 01:30 p.m. To 04:30 p.m.

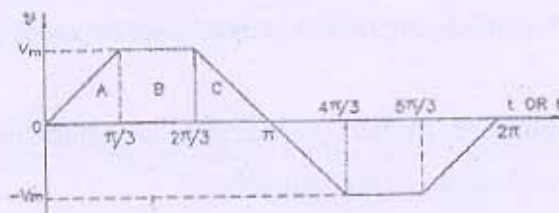
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Determine the average and RMS value of a trapezoidal waveform as shown in Figure. [5]
Also determine form factor and the crest factor.



- (b) Define Form factor and Crest factor. [2]

- Q - 2 (a) Explain resonance in series RLC circuit & its Q-factor. [7]

- (b) An inductive coil having an inductance of 0.04 H and a resistance of 5Ω is placed in parallel with a variable capacitor. The circuit is connected across a 100 V, 50 Hz supply. For what value of capacitor will the line current be minimum and what are the values of two branch currents, the total power and power factor of the circuit? [7]

OR

- Q - 2 (a) Give the similarities and dissimilarities between Electric and Magnetic circuits. [7]

- (b) Define the following terms: [7]

- (i) Magnetic flux density
- (ii) Magnetic field intensity
- (iii) Permeability
- (iv) Reluctance
- (v) Permeance

- Q – 3(a) Derive the relationship between line current & phase current in a 3-phase delta-connected circuit. [1]
- (b) Three similar coils each of resistance $15\ \Omega$ and inductance of $0.25\ \text{H}$ are connected (i) [7]
in star and (ii) in delta to a 3-phase, $400\ \text{V}$, $50\ \text{Hz}$ supply. Calculate line and phase values of current and voltage in both the cases. Also calculate the power absorbed.

OR

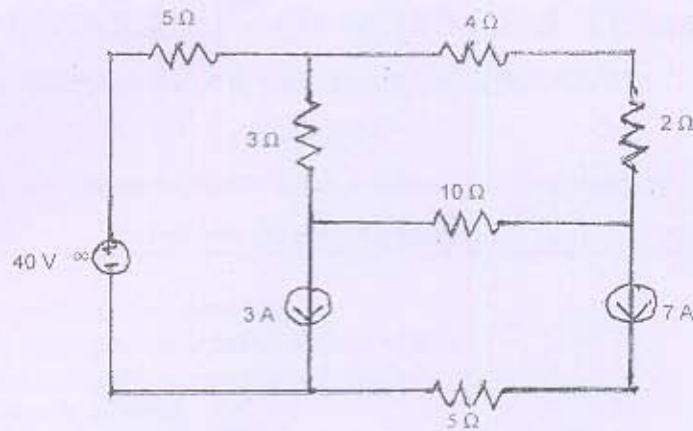
- Q – 3 (a) Explain working principle and application of transformer. [7]
- (b) Explain Basic working principle of Generator and its classification. [7]

SECTION – II

- Q – 4 (a) What do you understand by the term “resistance”? State and explain factors affecting resistance of a conductor. [5]
- (b) Convert $(125)_{10}$ into hexadecimal number. [2]
- Q – 5 (a) Why voltage stabilizer circuit is required? Explain zener diode working as a voltage stabilizer. [7]
- (b) Explain the common emitter configuration of transistor in detail. Derive current amplification and output current equations. [7]

OR

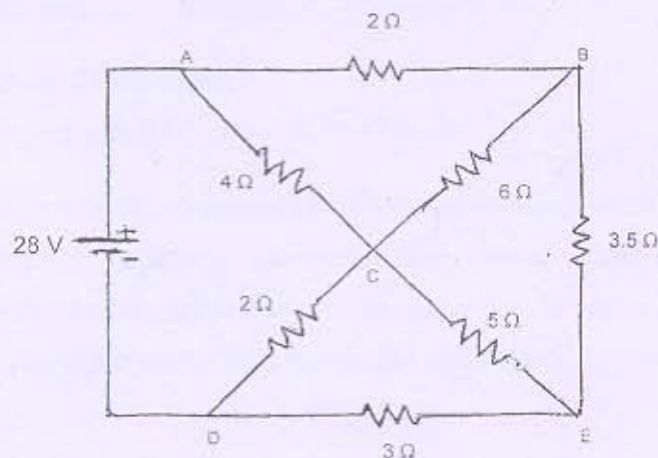
- Q – 5 (a) Derive an expression for [7]
1. Capacitance of parallel plate with partly air medium,
2. Multi-plate capacitor having N plates
- (b) A $2\ \mu\text{F}$ capacitor is connected by closing the switch to a supply of $100\ \text{V D.C.}$ through a $1\ \text{M}\Omega$ series resistance. Calculate (i) the time constant (ii) Initial charging current (iii) voltage across capacitor $6\ \text{sec}$ after the switch has been closed (iv) time taken for capacitor to be fully charged (v) circuit current at time equal to time constant (vi) time taken for the capacitor voltage to reach $70\ \text{volts}$ (vii) charge after $0.5\ \text{seconds}$. [7]
- Q – 6 (a) A crystal diode having internal resistance of $R_f = 20$ used in half wave rectification. If applied voltage $v = 60\sin\omega t$ and load resistance $R_L = 1000$. Find i) I_m , I_{dc} , I_{rms} , ii) AC input power and DC output power, iii) DC output voltage, iv) Rectifier Efficiency [4]
- (b) Explain the different types of filter circuit [4]
- (c) Determine the power delivered by the voltage source and current passing through $10\ \Omega$ resistor for the circuit shown in below figure. [6]



OR

- Q - 6 (a) Two conductors were carrying equal current at 20°C when connected in parallel. Find the ratio of currents passing through them if temperature is increased to 80°C . The temperature coefficient of resistances of conductors 1 & 2 are $\frac{1}{232.56}$ per $^{\circ}\text{C}$ and $\frac{1}{181.82}$ per $^{\circ}\text{C}$. [5]

- (b) Calculate current supplied by the battery for the given figure. [4]



- (c) Define Electric potential, Field Intensity and Flux density and give relationship between them. [5]
